Fault-Conditioned UAV World Modeling for State Prediction and Contingency Decision Support

***Abstract*—Post-fault unmanned aerial vehicle (UAV) safety requires anticipating vehicle evolution under degraded or unavailable observations to support informed contingency assessment. This study explores learned world modeling for fault-conditioned UAV prediction and emergency-response evaluation. We propose a memory-augmented structured world model (MAS-WM) that integrates fault-conditioned dynamics with memory-enhanced inference and a shared-parameter reference-posterior KL objective. By introducing an unaugmented reference posterior, the model aligns measurement-informed state estimation with observation-free prediction without an additional inference network. In capacity-matched experiments, MAS-WM reduces position and velocity reconstruction errors by 19.17% and 13.69%, respectively. Multi-step and fault-conditioned evaluations further examine predictive capability, while sensitivity analysis characterizes the influence of retrieved context on different state components. A downstream actor-critic evaluates four contingency modes, revealing trade-offs between intervention preferences, task retention, and maneuver demands; flight observations further demonstrate state-dependent appraisal during execution. These findings illustrate the potential of world models to anticipate fault-induced UAV behavior and inform contingency decisions, providing an initial basis for future safety-aware, fault-tolerant UAV autonomy.**

***Index Terms*—Unmanned aerial vehicle (UAV), world model, fault-conditioned prediction, memory-enhanced inference, contingency decision support.**

I. Introduction

Reliable Unmanned aerial vehicle (UAV) operation requires balancing task progress against changing flight conditions [1], [2]. Faults complicate this balance because an acceptable current state may evolve differently under continuation, hover, landing, or return. Reduced actuation, for example, can affect position holding and return maneuvers differently, while sensor anomalies alter motion through feedback control. Response appraisal therefore requires predicting how observed motion, fault conditions, and candidate maneuvers jointly determine subsequent flight behavior before their consequences are fully observed [3].

World models (WMs) learn state transitions and task signals for imagined action evaluation [4]. DreamerV3 combines recurrent dynamics with reward and continuation prediction [5], while DayDreamer extends this paradigm to physical robot learning [6]. Reviews of learned dynamics and robot WMs connect predictive representations to planning and policy learning [7]–[9]. Think2Drive, Epona, and World4Drive model action- or intention-conditioned driving futures [10]–[12]; MaskGWM and the Sora survey address visual prediction and generation [13], [14]. CTRL-WORLD and Chain of World explore controllable generation and latent motion reasoning [15], [16], RLVR-World investigates reinforcement learning for WM training [17], and WorldMem incorporates historical context into prediction [18]. These developments motivate the use of WMs for UAV response appraisal, where numerical flight-state prediction and task-signal prediction must jointly characterize the consequences of candidate responses under faults.

Applying WMs to this setting requires distinguishing the prediction task from fault recognition and response execution. Uncertainty studies and safety assessments characterize abnormal operating conditions [19], [20], while fault identification and learning-based response address their recognition and handling [21]–[25]. Safety-critical quadrotor control accounts for actuator faults and output constraints [26], and risk-aware predictive control supports resilient multi-UAV coordination [27]–[29]. Whereas these approaches emphasize constraint handling and response design, a learned conditional model addresses a complementary question: *how will the controlled vehicle evolve under a supplied fault condition and a candidate response?* Actuator faults change available actuation, whereas sensor faults distort the feedback used to regulate motion. Predicting their consequences therefore requires jointly representing the supplied fault condition, observed flight state, and candidate mode.

The inference-imagination gap. The effectiveness of imagined evaluation depends on how latent states are inferred and subsequently propagated. Uncertainty-aware model-based driving incorporates predictive uncertainty into decision-making [30]; a related concern in the present setting is the difference between information available during observation-conditioned inference and prior imagination. Measurements and retrieved transitions can refine the posterior, whereas prior rollouts proceed without these corrections. ***Improved reconstruction therefore does not necessarily translate into improved multi-step prediction.*** Under faults, the latent dynamics must additionally preserve the relationship between abnormal conditions and ensuing motion. Learning should thus enrich posterior inference while maintaining a predictive prior that remains effective after observation correction and memory retrieval cease.

To address this requirement, we propose a **memory-augmented structured world model (MAS-WM)**. Fig. 1 illustrates three connected stages: *(i) representation mapping* encodes flight and fault information alongside the candidate mode; *(ii) dynamic modeling and state inference* integrates recurrent motion context with flight/fault branches; and *(iii) generation and imagination rollout* predicts states, rewards, and continuation for mode appraisal. ***The specific learning contribution is a shared-parameter reference-posterior objec-***

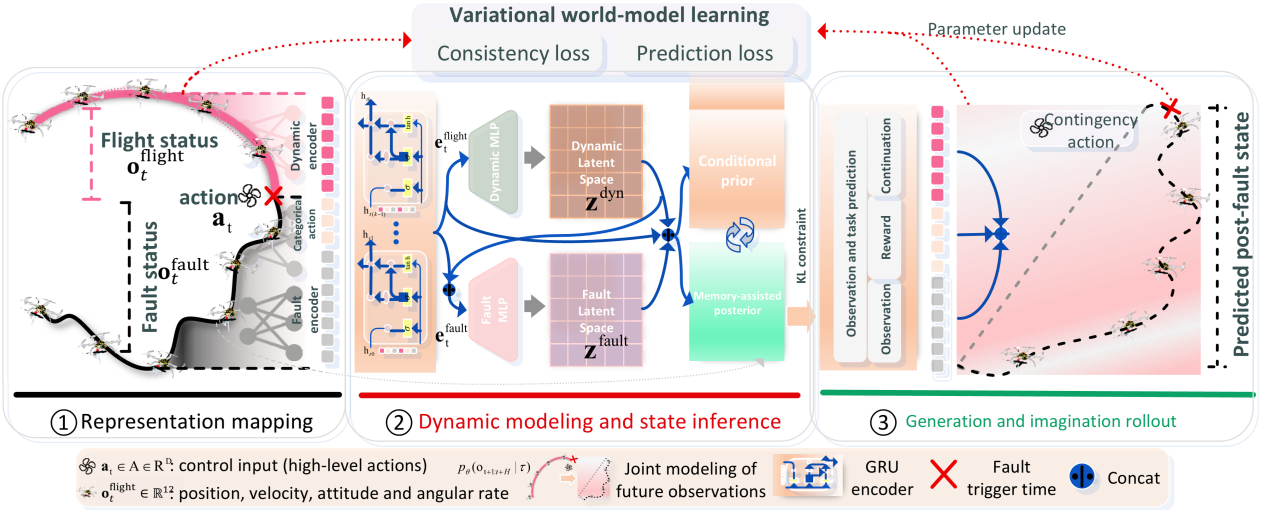


Fig. 1. MAS-WM framework. (1) Representation mapping: Section II-B. (2) Dynamic modeling and state inference: conditional dynamics in Section II-B and memory/reference-posterior learning in Section II-C. (3) Generation and imagination rollout: prediction heads in Section II-C.3, with the downstream contingency-appraisal application in Section III-E.1. Top consistency/prediction losses correspond to Sections II-C.2/II-C.3. Prior rollout uses no future observations or retrieval.

tive: a memory-enhanced posterior and a reference posterior that omits the current retrieval residual jointly regularize the same prior. Together with the prediction losses in Fig. 1, these complementary targets connect observation-conditioned inference to subsequent prediction without measurements or retrieval. Capacity-matched experiments evaluate the incremental effect of this objective, while contingency-appraisal experiments examine the downstream use of the learned predictions. Our contributions are:

1) ***Fault-conditioned UAV state modeling*** that connects structured flight/fault representations to observation-free prediction under candidate contingency modes.
2) ***A shared-parameter reference-posterior objective*** that complements memory-enhanced inference, with capacity-matched ablations separating posterior reconstruction from subsequent prior prediction.
3) ***A contingency-appraisal application*** that links predicted state and task signals to four high-level modes, evaluated through policy distributions, controlled mode responses, and flight observations.

II. Methodology and Framework

A. Prediction task and framework

As shown in Fig. 1, the inputs are the observation $\mathbf{o}_t = (\mathbf{o}_t^{\text{flight}}, \mathbf{f}_t)$ and mode $\mathbf{a}_t$. The flight vector $\mathbf{o}_t^{\text{flight}} \in \mathbb{R}^{12}$ comprises position, velocity, Euler angles, and angular rates, while $\mathbf{f}_t \in \mathbb{R}^2$ encodes fault type and severity (denoted $\mathbf{o}_t^{\text{fault}}$ in Figs. 1-2). The one-hot action $\mathbf{a}_t$ selects from $\mathcal{A} = \{\text{continue}, \text{hover}, \text{land}, \text{return}\}$. Given history $\mathcal{H}_t$ and an action sequence, the right-hand module predicts future flight states, rewards, and continuation labels $c = 1 - d$, where d denotes termination:

$$p_\theta(\mathbf{o}^{\text{flight}}_{t+1:t+H}, r_{t+1:t+H}, c_{t+1:t+H} \mid \mathcal{H}_t, \mathbf{a}_{t:t+H-1}, \mathbf{f}_t). \quad (1)$$

Here, H is the prediction horizon and θ denotes world-model parameters. Sections II-B-II-C detail latent-state construction and the transition from observation-conditioned inference to prior prediction. Section III-E.1 describes how these predictions support contingency appraisal.

Remark 1 (Fault-conditioned input) Fault detection and diagnosis identify anomaly type and severity, whereas this work predicts fault-conditioned motion under candidate responses to support emergency assessment. Fault conditions, distinct from flight observations, serve as inputs potentially provided by upstream modules. Experiments inject known faults individually and hold each condition fixed over the prediction horizon, isolating conditional-dynamics prediction from identification errors.

Remark 2 (High-level discrete action space) The four actions define a high-level discrete emergency-mode space, hierarchically decoupled from low-level continuous control. Each mode specifies the mission and motion references, while the existing flight controller generates commands and ensures closed-loop regulation. This separation enables response-conditioned motion prediction without learning continuous control laws or addressing their associated control-allocation, actuator-constraint, and stability complexities.

B. Structured representations and conditional dynamics

Fig. 2 details the first two stages of Fig. 1: Block (i) encodes observations and maintains recurrent history; Block (ii) propagates conditional priors; and Block (iii) refines posterior inference via retrieval and provides two posterior targets for prior learning. These mechanisms jointly support observation-conditioned inference, whereas imagination propagates the initialized recurrent state through the priors.

1) Encoding and recurrence: The encoders ψ_f and ψ_c in Fig. 2(i) map flight and fault inputs to the corresponding branches:

$$\mathbf{e}_t^{\text{flight}} = \psi_f(\mathbf{o}_t^{\text{flight}}), \quad \mathbf{e}_t^{\text{fault}} = \psi_c(\mathbf{f}_t), \\ \mathbf{z}_t = [\mathbf{z}_t^{\text{dyn}}; \mathbf{z}_t^{\text{fault}}]. \quad (2)$$

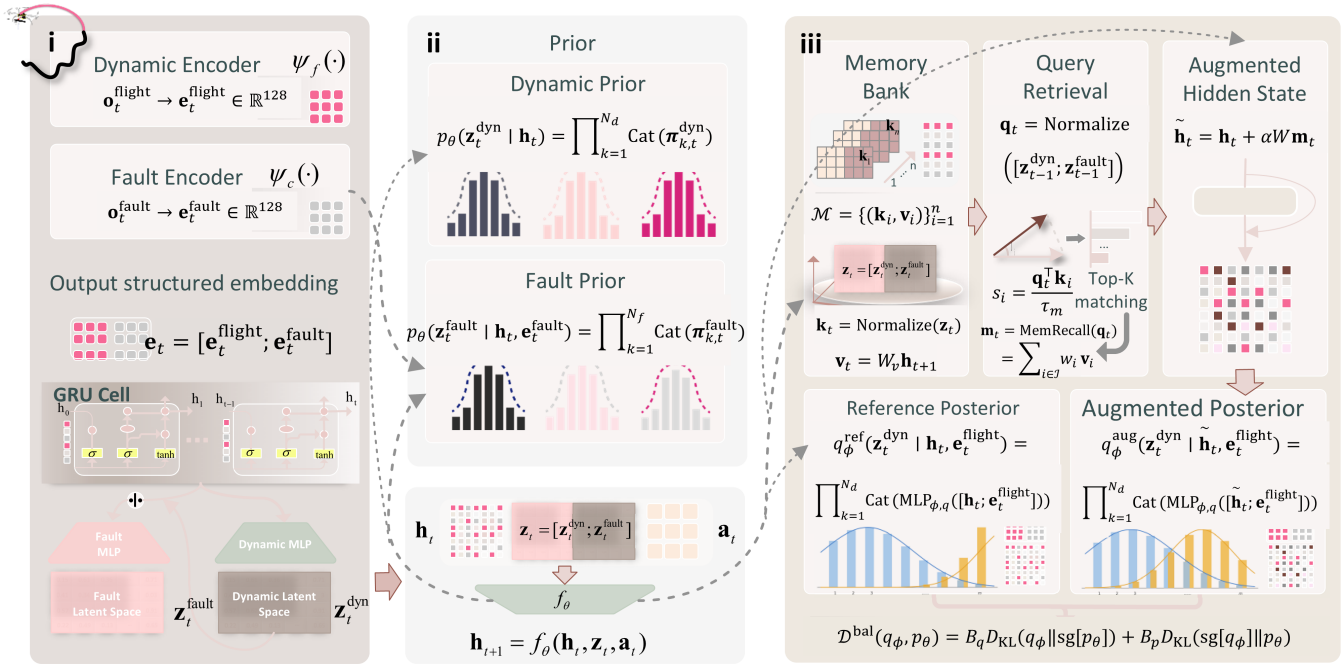


Fig. 2. Inference and prediction mechanisms. (i) Dual encoders and shared recurrence, Section II-B. (ii) Conditional priors, Section II-B. (iii) Top-K retrieval, Section II-C.1, and augmented/reference posteriors regularizing the same prior, Section II-C.2. Fault posteriors follow the illustrated dynamics-posterior structure; retrieval operates only during observation-conditioned inference.

The latent state combines a deterministic recurrent component $\mathbf{h}_t$ with components $\mathbf{z}_t$:

$$\mathbf{h}_t = f_\theta(\mathbf{h}_{t-1}, \mathbf{z}_{t-1}, \mathbf{a}_{t-1}), \qquad \mathbf{s}_t = [\mathbf{h}_t; \mathbf{z}_t]. \tag{3}$$

2) Conditional prior propagation: The transition f_θ retains temporal dependence and previous actions. Each branch consists of categorical factors, with N_d and N_f factors for dynamics and faults. Their priors in Fig. 2(ii) are

$$\begin{aligned} p_\theta(\mathbf{z}_t^{\text{dyn}} \mid \mathbf{h}_t) &= \prod_{k=1}^{N_d} \text{Cat}(\boldsymbol{\pi}_{k,t}^{\text{dyn}}), \\ p_\theta(\mathbf{z}_t^{\text{fault}} \mid \mathbf{h}_t, \mathbf{e}_t^{\text{fault}}) &= \prod_{k=1}^{N_f} \text{Cat}(\boldsymbol{\pi}_{k,t}^{\text{fault}}). \end{aligned} \tag{4}$$

$\boldsymbol{\pi}_{k,t}^b$ is the probability vector for factor k of branch b, produced by a branch-specific multilayer perceptron. Straight-through Gumbel-Softmax sampling trains the categorical states. The dynamics prior receives recurrent motion context, and the fault prior additionally receives the fault embedding. Once initialized by the posterior, these priors and Eq. (3) generate future states without new measurements. This distinguishes the transition's role from the observation correction introduced next.

C. Memory-enhanced inference and reference-posterior learning

1) Retrieval-conditioned posterior: The observation-stage path in Fig. 2(iii) complements prior propagation with a current-state estimate through four steps: *query, retrieve, augment, infer*. The previous latent state queries related transitions, whose retrieved values form a recurrent residual that augments the posterior input. The memory bank stores normalized latent keys and projected subsequent-state values:

$$\begin{aligned} \mathcal{M} &= \{(\mathbf{k}_i, \mathbf{v}_i)\}_{i=1}^{n}, \quad n \le C, \\ \mathbf{k}_t &= \mathbf{z}_t / \|\mathbf{z}_t\|_2, \qquad \mathbf{v}_t = W_v \mathbf{h}_{t+1}. \end{aligned} \tag{5}$$

Here, C denotes capacity and W_v projects recurrent states into the value space. Entries are written after model updates without gradient propagation. A query from the previous latent state selects similar transitions:

$$\begin{aligned} &\mathbf{q}_t = \mathbf{z}_{t-1}/\|\mathbf{z}_{t-1}\|_2, \quad s_i = \mathbf{q}_t^\top \mathbf{k}_i / \tau_m, \\ &\mathbf{m}_t = \sum_{i \in \mathcal{I}_K} \frac{\exp(s_i)}{\sum_{j \in \mathcal{I}_K} \exp(s_j)} \mathbf{v}_i, \quad \tilde{\mathbf{h}}_t = \mathbf{h}_t + \alpha W \mathbf{m}_t. \end{aligned} \tag{6}$$

$\mathcal{I}_K$ contains the K highest-similarity keys, $\tau_m > 0$ controls retrieval concentration, W maps values to the recurrent space, and $\alpha \ge 0$ controls residual strength. For $b \in \{\text{dyn}, \text{fault}\}$, the augmented and reference posteriors share parameters ϕ but receive different recurrent inputs:

$$q_\phi^{\text{aug},b}(\mathbf{z}_t^b \mid \tilde{\mathbf{h}}_t, \mathbf{e}_t^b), \qquad q_\phi^{\text{ref},b}(\mathbf{z}_t^b \mid \mathbf{h}_t, \mathbf{e}_t^b). \tag{7}$$

Here $\mathbf{e}_t^{\text{dyn}} = \mathbf{e}_t^{\text{flight}}$. The augmented posterior uses retrieved context to estimate the observed state; the reference posterior preserves a target without that residual. ***Their shared parameters make the subsequent comparison a change in conditioning, not an additional inference network.***

2) Reference-posterior regularization: The two posterior paths in Fig. 2(iii) connect retrieval to prior learning. The augmented path supplies a context-enriched target; the reference path uses the same network without the current residual. Balanced KL divergence controls gradients to posterior and prior separately:

$$\mathcal{D}^{\text{bal}}(q, p) = B_q D_{\text{KL}}(q \| \,\text{sg}[p]) + B_p D_{\text{KL}}(\text{sg}[q] \| p). \tag{8}$$

TABLE I
STATE RECONSTRUCTION AND TASK-SIGNAL PREDICTION OF MAS-WM AND BASELINES.

Model	pos RMSE (m) ↓	vel RMSE (m/s) ↓	att error (°) ↓	rew MAE – ↓	rew RMSE – ↓	KL –	AUC – ↑	recon L2 – ↓
DreamerV3	0.2891	0.3398	3.2944	0.0359	0.0653	0.2083	0.9457	0.4586
CoWVLA	0.3102	**0.2906**	3.1124	0.0354	0.0660	0.6069	0.9332	0.4282
DayDreamer	0.4058	0.3486	3.5363	3.5166	3.6288	0.1728	0.9132	0.5396
CTRL-WORLD	1.0875	0.6581	6.0955	0.0487	0.0844	0.5935	0.5494	0.8576
LoGeR	9.0885	1.1945	21.3121	–	–	–	–	–
RISE	**0.1346**	0.4081	16.3068	–	–	–	–	1.3184
MAS-WM	0.2431	0.2909	**2.9533**	**0.0294**	**0.0556**	0.2706	**0.9638**	**0.3841**

Note: Recurrent baseline state results are posterior reconstructions over fifty episodes. Metrics and aggregation follow Section III-B.3; position/velocity use vector RMSE, attitude uses mean Euclidean error in degrees, and recon. L2 is the mean observation-residual norm. Baselines with different output structures retain their reported estimates. AUC measures recorded continuation-label discrimination. Tint identifies MAS-WM; bold marks minimum errors or maximum AUC.

TABLE II
ARCHITECTURE COMPARISONS AND CAPACITY-MATCHED INFERENCE/OBJECTIVE ABLATIONS.

Model	pos RMSE (m) ↓	vel RMSE (m/s) ↓	att error (°) ↓	rew MAE – ↓	rew RMSE – ↓	KL –	AUC – ↑	recon L2 – ↓
Architecture-level ablations								
ActionEmbed* (w/o fault paths)	0.2759	0.3075	3.0396	0.0345	0.0634	0.2109	0.9449	0.4238
FaultAware† (w/o split states)	0.2778	0.2910	3.0429	0.0334	0.0598	0.1906	0.9577	0.4082
SplitLatent‡ (w/o memory/ref. KL)	0.2834	0.3065	3.0566	**0.0292**	0.0563	0.2594	0.9587	0.4157
Capacity-matched inference/objective ablations								
w/o memory and ref. KL	0.3008	0.3370	3.3605	0.0349	0.0639	0.2414	0.9510	0.4455
w/o ref. KL	0.2772	0.3104	3.1484	0.0305	0.0581	0.2324	0.9603	0.4150
w/o memory (ref. KL control)	0.2807	0.3275	3.0402	0.0313	0.0597	0.2365	0.9605	0.4155
MAS-WM	**0.2431**	**0.2909**	**2.9533**	0.0294	**0.0556**	0.2706	**0.9686**	**0.3841**

Note: *ActionEmbed changes action inputs without fault branches; †FaultAware adds fault conditioning to an unsplit state; ‡SplitLatent separates branches with shared recurrence (Section II-B), without retrieval/reference KL. These are noncumulative comparisons. The lower block fixes 11495952 parameters, 5767168 memory elements, data, seed, batch/sequence sizes, updates, and 50 evaluation episodes. Its three w/o settings use $(\alpha, w) = (0,0), (0.05,0), (0,0.2)$; MAS-WM uses $(0.05, 0.2)$. Memory changes the posterior input (Section II-C.1); reference KL adds the unaugmented-posterior target (Section II-C.2). Metrics follow Section III-B.3. Bold marks optima except KL. AUC 0.9686 re-evaluates the same MAS-WM weights as Table I's 0.9638; the difference is not a training gain.

$B_q, B_p \geq 0$ with $B_q + B_p = 1$ are gradient weights; sg stops differentiation through its argument. The branch-weighted objective combines both posterior targets:

$$\mathcal{L}_{\mathrm{KL}} = \sum_b \beta_b \left[(1-w)\mathcal{D}^{\mathrm{bal}}(q^{\mathrm{aug},b}, p^b) + w\mathcal{D}^{\mathrm{bal}}(q^{\mathrm{ref},b}, p^b) \right]. \quad (9)$$

The branch weight β_b and $w \in [0,1]$ govern branch contributions and reference regularization, respectively. The first term aligns the prior with memory-enhanced inference, while the second regularizes it toward the unaugmented-state posterior, preventing exclusive reliance on retrieval-enhanced targets. Both terms constrain the same prior. Section III-C.3 evaluates whether this dual-target objective improves prediction when retrieval is unavailable.

Remark 3 (Representation and objective) The branches share recurrence and do not imply statistical independence or causal disentanglement. The reference omits the current retrieval residual, not historical memory effects carried by recurrence. Reference regularization adds no stability guarantee. For $\alpha = 0$, the posteriors coincide and Eq. (9) is independent of w; reference-KL effects must therefore be identified with memory enabled.

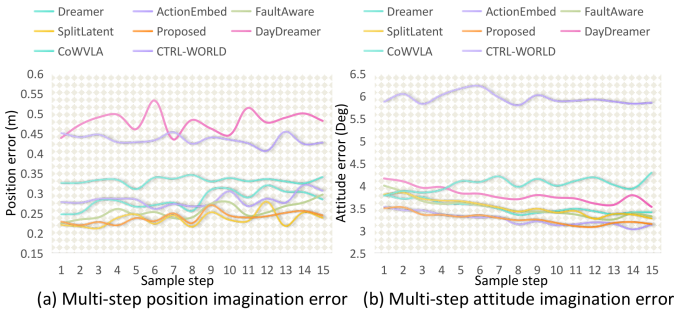
3) State and task prediction heads: The generation module in Fig. 1 decodes inferred and imagined states into observations and task signals:

$$p_\theta(\mathbf{o}_t \mid \mathbf{s}_t), \qquad p_\theta(r_t \mid \mathbf{s}_t), \qquad p_\theta(c_t \mid \mathbf{s}_t). \quad (10)$$

Joint optimization minimizes

$$\mathcal{L}_{\mathrm{WM}} = \mathcal{L}_{\mathrm{obs}} + \mathcal{L}_r + \mathcal{L}_c + \lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}}. \quad (11)$$

Here, $\mathcal{L}_{\mathrm{obs}}$ is masked observation MSE, $\mathcal{L}_r$ and $\mathcal{L}_c$ fit rewards and continuation labels, respectively, and λ_{KL} controls regularization. Temporal masks exclude padding, and a free threshold limits KL penalties. As summarized above Fig. 1, the shared heads jointly support consistency learning and prediction, decoding posterior states during observation and



(a) Multi-step position imagination error (b) Multi-step attitude imagination error

Fig. 3. Multi-step position and attitude prediction from posterior-initialized states. The prior propagates posterior-initialized states for fifteen steps without future observations.

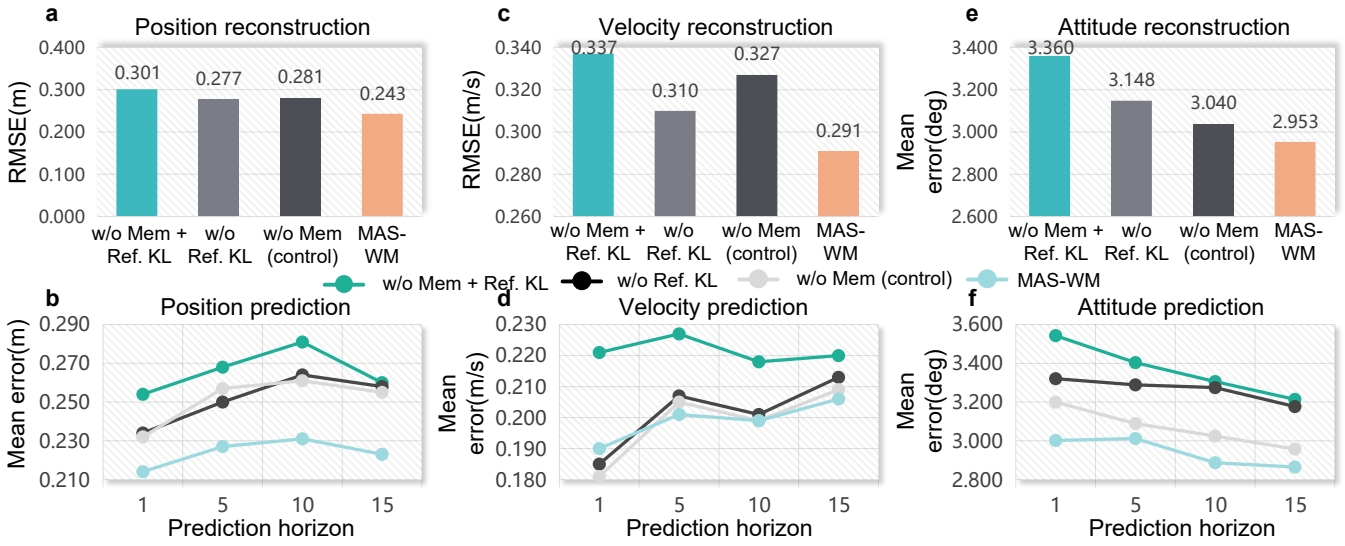


Fig. 4. Prediction-level complement to Table II. (a,c,e) Posterior reconstruction. (b,d,f) Mean Euclidean prior errors at 1, 5, 10, and 15 steps after retrieval stops. Memory changes the posterior residual (Section II-C.1); Ref. KL changes its training target (Section II-C.2).

prior states during imagination. This common output space enables downstream evaluation of candidate responses.

III. Experiments and Verification

A. Validation objectives

Following the inference-prediction-appraisal sequence in Fig. 1, experiments address three research questions:

- ***RQ1: Does enhanced inference improve subsequent prediction?*** Section III-C evaluates reconstruction, observation-free rollout, matched-objective ablations, and retrieval sensitivity (Sections II-B-II-C).
- ***RQ2: Can the model predict motion consequences under specific faults?*** Section III-D evaluates conditional dynamics across fault types and severities (Section II-B).
- ***RQ3: How does the learned representation support contingency appraisal?*** Sections III-E-III-F examine mode distributions, controlled response outcomes, and flight observations.

B. Experimental setup

1) Data and baselines: `CopterSim`[1] and `RflySim3D`[2] provide vehicle dynamics and fault injection. The dataset contains 499 flight episodes with motor, accelerometer, and gyroscope faults; reconstruction and prediction evaluations use episodes drawn from this dataset. Baselines are `DreamerV3`, `DayDreamer`, `CoWVLA`, `CTRL-WORLD`, `LoGeR`, and `RISE` [5], [6], [15], [16], [31], [32]. To compare prediction of the same physical quantities under the same mode definitions, their task interfaces use numerical UAV observations and four categorical modes in place of the original inputs and actions.

[1] https://rflysim.com/doc/en/3.Soft/coptersim.html
[2] https://rflysim.com/doc/en/3.Soft/3d-eng.html

2) Training and execution settings: The recurrent state has 512 units; the dynamics and fault branches use 48 and 32 factors, respectively, each with 32 categories. Encoders and prediction heads comprise two 256-unit hidden layers. Offline WM training uses seed 42, batch size 16, sequence length 64, and learning rate 10^{-4}. The settings in Eqs. (8)-(11) are $(B_q, B_p) = (0.6, 0.4)$, $(\beta_{\text{dyn}}, \beta_{\text{fault}}, w) = (1, 2, 0.2)$, $\lambda_{\text{KL}} = 1$, and a free threshold of 0.1. Memory defaults are $(C, K, \tau_m, \alpha) = (2048, 8, 0.1, 0.05)$ with 256-dimensional values. Recurrent states reset each episode, whereas memory persists; prior prediction neither retrieves nor writes to memory. At 20 Hz, 15- and 40-step horizons correspond to 0.75 and 2 s, respectively.

3) Evaluation metrics: For residual $\mathbf{e}_i = \hat{\mathbf{x}}_i - \mathbf{x}_i$ over N valid samples, evaluation metrics are grouped as follows.

- ***State prediction and reconstruction.*** Tables I and II report vector RMSE for position/velocity and mean Euclidean error (ME) for Euler angles; multi-step component comparisons use ME, while fault-group and sensitivity analyses use coordinate RMSE:

$$\begin{aligned} E_{\text{vec}} &= \sqrt{N^{-1}\textstyle\sum_i \|\mathbf{e}_i\|_2^2}, \\ E_{\text{ME}} &= N^{-1}\textstyle\sum_i \|\mathbf{e}_i\|_2, \\ E_{\text{coord}} &= \sqrt{(ND)^{-1}\textstyle\sum_i \|\mathbf{e}_i\|_2^2}. \end{aligned} \tag{12}$$

  Here, D is the vector dimension. Reconstruction L2 is E_{ME} on observation residuals, and reconstruction MSE is E^2_{coord}. State residuals in Tables I-II are pooled across 50 episodes; reconstruction summaries average episode statistics. Sensitivity results average per-episode coordinate RMSE.
- ***Reward prediction.*** For reward residual e^r_i, mean absolute

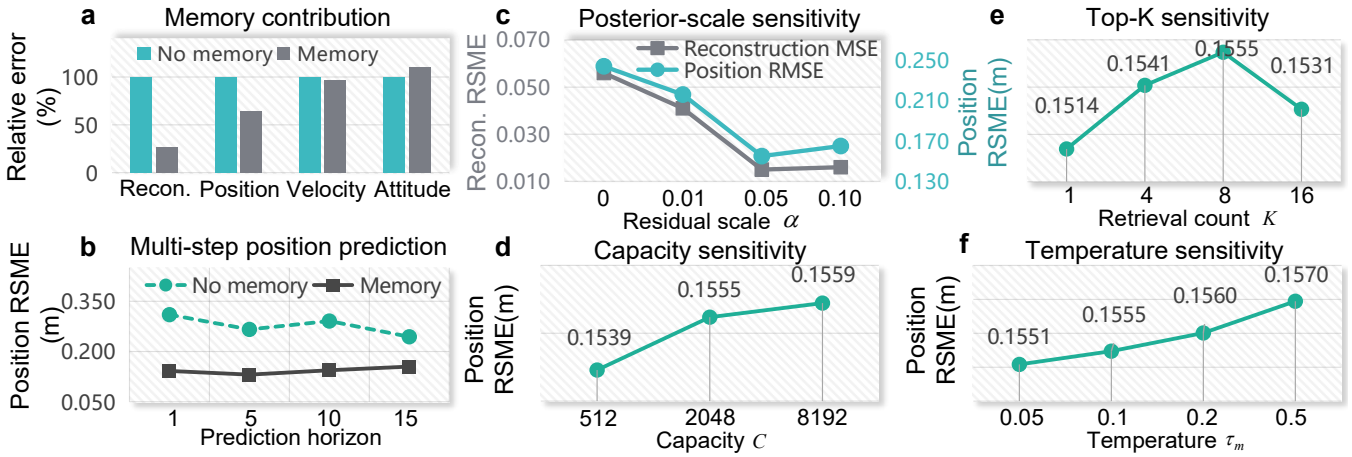


Fig. 5. Sensitivity to (α, C, K, τ_m) in Eq. (6). (a) Errors relative to no memory (100%). (b) Prior prediction. (c) Residual strength α. (d)-(f) Capacity C, neighbor count K, and temperature τ_m; peach markers denote defaults. Sixteen fixed-weight settings rebuild memory from twenty episodes and evaluate ten disjoint episodes of the same dataset; memory is frozen unless write/reset is explicitly varied.

TABLE III
FAULT-CONDITIONED FORTY-STEP PRIOR PREDICTION ERRORS.

Method	pos RMSE (m) ↓				vel RMSE (m/s) ↓				att RMSE (rad) ↓			
	Motor	Accel	Gyro	Avg.	Motor	Accel	Gyro	Avg.	Motor	Accel	Gyro	Avg.
MAS-WM	**0.1328**	**0.0593**	0.2645	**0.1522**	0.0502	**0.0899**	0.1903	0.1101	**0.0138**	**0.0185**	**0.0190**	**0.0171**
FaultAware	0.1646	0.1415	0.4247	0.2436	**0.0389**	0.1061	0.1390	**0.0947**	0.0206	0.0326	0.0293	0.0275
DayDreamer	0.1981	0.3130	**0.2612**	0.2574	0.1314	0.1002	0.1296	0.1204	0.0175	0.0417	0.0466	0.0353
DreamerV3	0.2184	0.3634	0.4616	0.3478	0.1081	0.0995	0.1552	0.1209	0.0192	0.0259	0.0323	0.0258
CoWVLA	0.2298	0.2611	0.3000	0.2636	0.1370	0.1640	**0.0922**	0.1311	0.0217	0.0352	0.0272	0.0280
CTRL-WORLD	0.2578	0.4921	0.2944	0.3481	0.2759	0.4501	0.2159	0.3140	0.0676	0.0497	0.0550	0.0574

Note: Thirty observed steps initialize forty prior steps (1.5/2.0 s at nominal 20 Hz). Coordinate RMSE in Eq. (12) aggregates residuals within each fault group; Avg. averages motor, accelerometer, and gyroscope groups. Position, velocity, and attitude use m, m/s, and rad. Bold marks column minima.

error (MAE) and root mean square error (RMSE) are

$$\begin{aligned} \mathrm{MAE}_r &= N^{-1} \sum_i |e_i^r|, \\ \mathrm{RMSE}_r &= \sqrt{N^{-1} \sum_i (e_i^r)^2}. \end{aligned} \tag{13}$$

- ***Latent regularization and continuation prediction.*** KL measures posterior-prior divergence rather than prediction error; its summaries average episode statistics. Continuation discrimination is evaluated by AUC:

$$\mathrm{AUC} = \Pr(\hat{c}^+ > \hat{c}^-) + \tfrac{1}{2} \Pr(\hat{c}^+ = \hat{c}^-), \tag{14}$$

where $\hat{c}^+$ and $\hat{c}^-$ denote predicted continuation scores for recorded positive and negative labels, respectively.

- ***Policy-distribution diagnostics.*** For severity bin j, let $\bar{\pi}_j$ denote the mean action distribution. Entropy, active action count (AAC), and policy specialization index (PSI) are

$$\begin{aligned} H_j &= -\sum_{a \in \mathcal{A}} \bar{\pi}_{j,a} \log_2 \bar{\pi}_{j,a}, \\ \mathrm{AAC}_j &= \sum_a \mathbb{I}[\bar{\pi}_{j,a} > 0.05], \\ \mathrm{PSI} &= 1 - \bar{H} / \log_2 |\mathcal{A}|. \end{aligned} \tag{15}$$

Here, $\bar{H}$ and mean AAC are averaged over valid bins. Phase entropy instead averages per-state entropy and reports its standard deviation. These metrics characterize action-probability concentration and the effective breadth of action support.

- ***Controlled response outcomes.*** The violation fraction and task retention are $\nu = N_{\mathrm{post}}^{-1} \sum_i \mathbb{I}[\text{any threshold exceeded}]$, $R = \min(S_{\mathrm{fault}}/S_{\mathrm{nominal}}, 1)$, where N_{post} is the number of post-response samples and $S = 0.7S_{\mathrm{track}} + 0.3S_{\mathrm{prog}}$ combines tracking and progress scores. Thresholds and crash definitions are specified alongside Table V.

C. Prediction accuracy and learning mechanisms

1) State reconstruction and task-signal prediction: Table I evaluates whether the inferred state preserves physical and task-relevant information for the prediction heads in Section II-C.3. Relative to `DreamerV3`, `MAS-WM` jointly reduces position, attitude, reward, and observation-residual errors, with velocity error close to `CoWVLA`'s minimum. In contrast, `RISE`'s lower position error accompanies substantially higher attitude error, highlighting the need to preserve multiple motion components for consequence appraisal. `MAS-WM`'s reward RMSE of 0.0556 and continuation AUC of 0.9638 further demonstrate retention of task-related signals beyond motion reconstruction. The shared state and task heads encode physical evolution and its task consequences, while KL regularization constrains posterior-prior agreement. Although low recon-

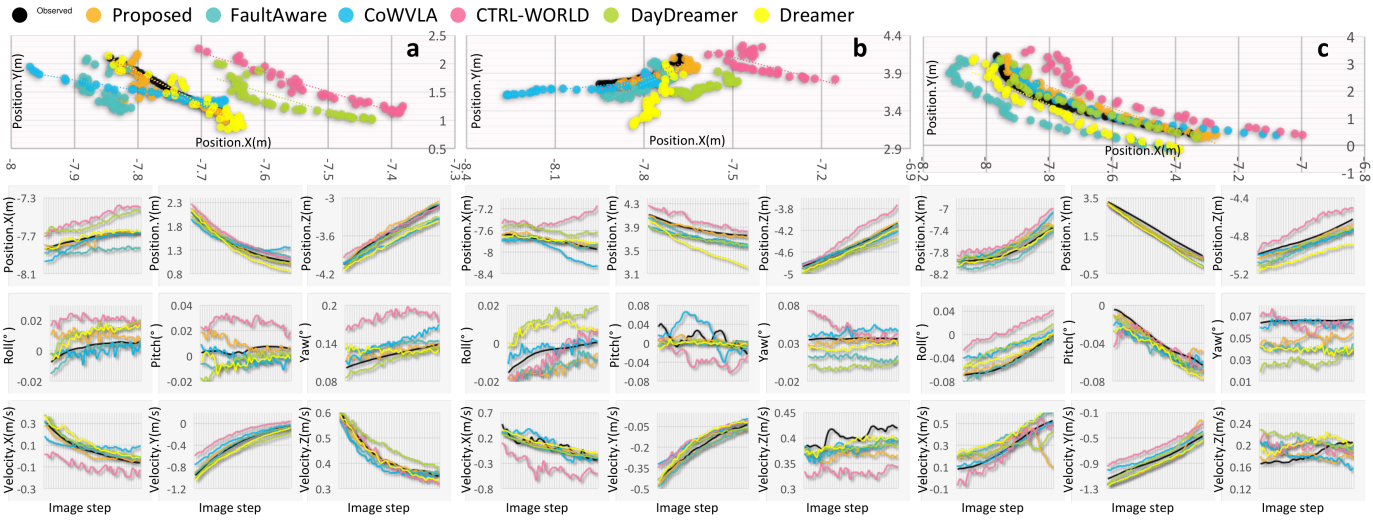


Fig. 6. Trajectory and state-trace comparison complementing Table III: (a) motor, (b) accelerometer, and (c) gyroscope faults. Recorded and conditional prior trajectories depict the controlled vehicle's response under the supplied fault condition, including spatial shape and component evolution.

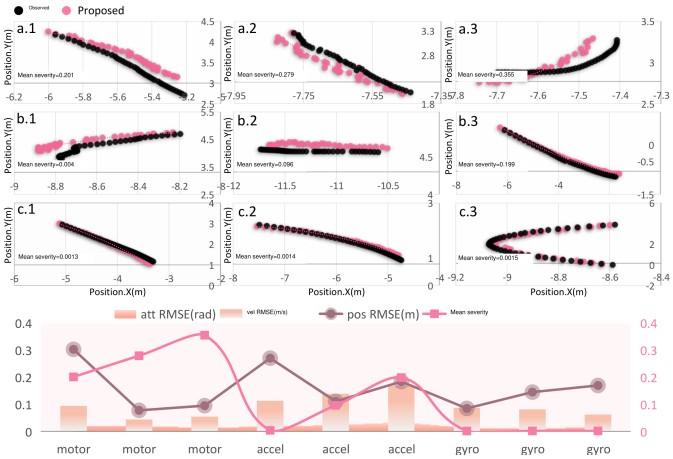


Fig. 7. Recorded and predicted trajectories grouped by supplied severity, extending Fig. 6. Bottom panels report component errors. Labels specify injected conditions, not diagnosed severity or a common physical scale across fault types.

struction and task-prediction errors support accurate observed-state inference, the prior-rollout and matched-ablation results separately assess whether this information remains predictive without observation correction.

2) Multi-step prior prediction: Accurate state and task-signal fitting initializes imagined trajectories. Fig. 3 evaluates observation-free position and attitude prediction, with `MAS-WM` remaining in the low-error group alongside SplitLatent for position and ActionEmbed for attitude. `DayDreamer` exhibits larger position errors, whereas `CTRL-WORLD` has higher attitude errors. Although error curves cross and vary nonmonotonically, `MAS-WM` maintains comparatively low errors across both components over fifteen prior steps. Without observation correction, trajectory accuracy depends jointly on initialization and learned dynamics; these results, together with Table I, indicate that `MAS-WM` preserves motion information useful for predicting both translation and orientation beyond observed-state reconstruction.

3) Ablation study and structural contributions: ***Posterior reconstruction.*** Table II shows complementary architectural effects: FaultAware achieves low velocity error, while SplitLatent yields the lowest reward MAE. Matched comparisons isolate the contributions of retrieval and the reference target in Fig. 2(iii). Memory reduces all three state errors, and reference regularization further improves reconstruction, lowering position and velocity errors by 19.17% and 13.69%, respectively, relative to the unaugmented model. Thus, the reference target preserves retrieval's reconstruction gains while introducing an additional objective for prior learning.

Prediction after retrieval stops. Fig. 4 separates reconstruction (a,c,e) from prior rollout (b,d,f). At 15 steps, memory alone yields a position error of 0.2585 m, close to 0.2600 m without memory, indicating limited predictive benefit at this horizon despite improved reconstruction. Adding the reference target reduces the memory-enabled position error to 0.2227 m, with velocity and attitude errors of 0.2059 m/s and 2.8652°. Jointly training against both posterior targets therefore helps preserve motion information for rollout after observation correction and retrieval cease.

The key reference-KL comparison is between the memory-only model and `MAS-WM`; at $\alpha = 0$, the two targets coincide. Gains vary with horizon and component, with memory alone slightly outperforming `MAS-WM` in one-step velocity prediction. Overall, Table II and Fig. 4 show complementary roles: ***retrieval improves observed-state inference, while reference regularization strengthens subsequent prediction***.

4) Memory parameter sensitivity: Fig. 5 evaluates retrieval parameters in Eq. (6) with fixed model weights. The residual strength α controls the influence of retrieved context: increasing it from zero to 0.05 improves position reconstruction and 15-step prediction, with smaller position gains at 0.1 (Fig. 5(c)). At the default setting, reconstruction MSE, velocity prediction error, and attitude error are 27.01%, 96.68%, and 109.84% of their no-memory values, respectively (Fig. 5(a)), revealing a trade-off that favors position accuracy over attitude accuracy. Capacity C, neighbor count K, and tempera-

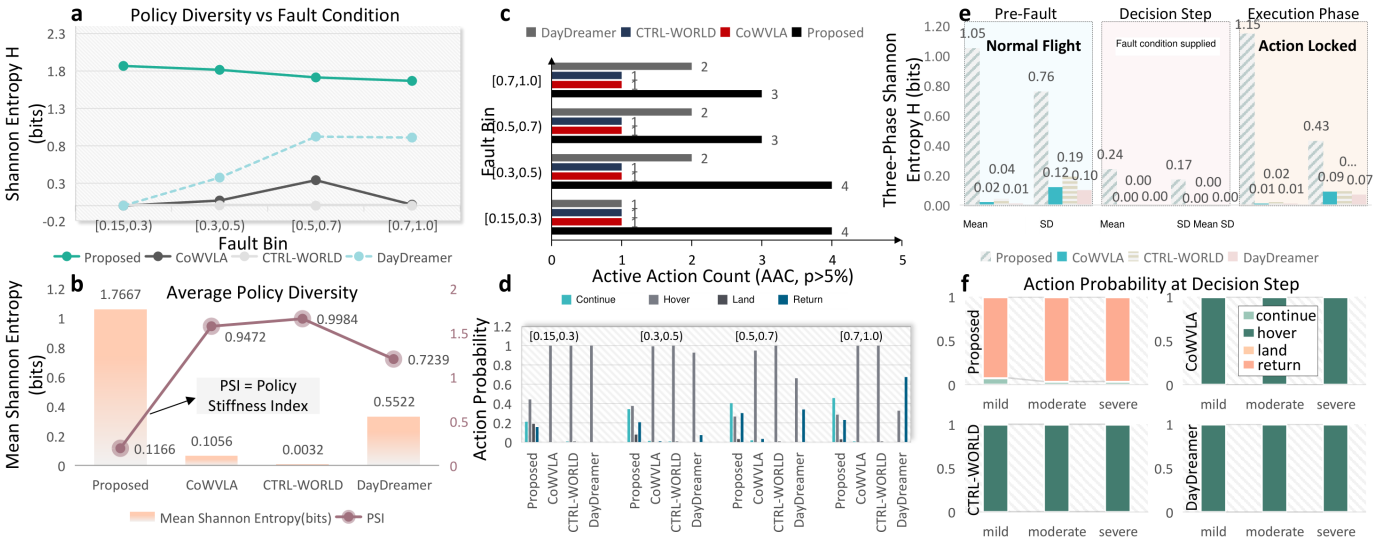


Fig. 8. Mode-appraisal diagnostics, summarized in Table IV. (a) Entropy of severity-bin mean probabilities. (b) Aggregate entropy/PSI (numerical values follow the table). (c) Active action count. (d) Bin-mean probabilities. (e) Phase-specific per-state entropy mean/SD. (f) Decision-step probabilities. Post-selection distributions evaluate evolving states while the command stays fixed.

TABLE IV
SEVERITY-BIN AND PHASE-SPECIFIC MODE-APPRAISAL STATISTICS.

Model	H (bits)	PSI	Mean AAC	H@Pre-fault	H@Decision	H@Execution	p_{Cont}@Exec
MAS-WM	1.767	0.117	3.50	1.054 ± 0.764	0.245 ± 0.169	1.148 ± 0.428	0.3093
CoWVLA	0.106	0.947	1.00	0.023 ± 0.120	0.000 ± 0.000	0.011 ± 0.085	0.0173
CTRL-WORLD	0.003	0.998	1.00	0.039 ± 0.193	0.004 ± 0.001	0.018 ± 0.088	0.0033
DayDreamer	0.552	0.724	1.75	0.014 ± 0.097	0.000 ± 0.000	0.008 ± 0.065	0.0000

Note: Definitions follow Eq. (15). H and AAC average severity-bin statistics; PSI is derived from H. Phase H is per-state entropy mean $\pm$ SD (bits), not entropy of the mean distribution. p_{Cont}@Exec is an evaluated probability under a fixed response mode. Numerical aggregates for Fig. 8 follow this table.

ture τ_m govern transition availability and retrieval weighting (Fig. 2(iii)). Across tested settings, 15-step position RMSE remains at 0.1514-0.1569 m, consistently below the 0.2436 m no-memory baseline (Fig. 5(d-f)), as further illustrated by prior-rollout trajectories in (Fig. 5(b)). Combined with the matched ablation, these results indicate that residual strength governs component-wise trade-offs, whereas reference regularization strengthens longer-horizon prediction.

D. Fault-conditioned prediction

Given a fault-affected flight segment, `MAS-WM` predicts the subsequent two seconds of position, velocity, and attitude without further measurements. Table III shows that its performance varies across fault types. Under motor degradation, it achieves the lowest position and attitude RMSEs (0.1328 m and 0.0138 rad); under accelerometer faults, it yields the lowest errors across all three components (0.0593 m, 0.0899 m/s, and 0.0185 rad). Under gyroscope faults, its attitude RMSE remains lowest (0.0190 rad), while its position error (0.2645 m) is comparable to `DayDreamer`'s (0.2612 m) and its velocity error exceeds several baselines. Thus, the lowest average position and attitude errors indicate robust prediction across fault conditions, although velocity accuracy remains more variable. Fig. 6 visualizes these responses. Under motor degradation (a), predicted trajectories capture the motion and attitude changes induced by reduced actuation; under accelerometer and gyroscope faults (b,c), they reproduce vehicle responses to disturbed feedback. Conditioning on fault type alongside recurrent motion and action information enables `MAS-WM` to distinguish these fault-dependent dynamics.

Fig. 7 further evaluates prediction across severity groups. Predicted paths preserve the main trajectory geometry, while the error panels quantify residual discrepancies. Position errors under motor and accelerometer faults decrease initially and then increase with severity, indicating that prediction difficulty depends not only on fault magnitude but also on the resulting controlled motion. Overall, Table III and Figs. 6-7 demonstrate that the fault-conditioned prior preserves useful position and attitude information over the two-second prediction horizon, despite heterogeneous errors across fault types and severity levels.

E. Downstream application cases: mode appraisal and controlled responses

We evaluate the predictive representation in an imagination-based actor-critic framework, where actions are assessed via predicted rewards and future states [5], [6]. Mode distributions and controlled trajectories characterize how learned dynamics support task retention under fault-induced maneuver constraints.

TABLE V
SELECTED MODES AND PHYSICAL OUTCOMES IN NINE CONTROLLED SCENARIOS.

Quantity	MAS-WM	CoWVLA	CTRL-WORLD	DayDreamer
Return / hover selections	9/0	0/9	0/9	0/9
Preserved feasible tasks	0/6	0/6	0/6	0/6
Interventions in feasible cases	6/6	6/6	6/6	6/6
Continued in severe cases	0/3	0/3	0/3	0/3
Severe cases avoiding crash	0/3	0/3	0/3	0/3
Crashes across selected branches	3/9	3/9	3/9	3/9
Mean violation fraction	0.358	0.296	0.296	0.296
Mean maximum attitude (degrees)	54.02	34.31	34.31	34.31
Mean maximum vertical speed (m/s)	3.214	2.639	2.639	2.639

Note: Outcomes use the forced branch matching each choice. Violation thresholds are absolute roll/pitch 30°, vertical speed 2 m/s, and NED ground position 0.3 m (intended landing contact excluded). Crash is attitude $> 50°$ or ground contact without safe landing; safe touchdown requires $|v_z| \leq 1$ m/s and attitude $\leq 15°$. In Section III-B.3, $S_{\text{track}} = 1 - \text{mean}\,\text{clip}(e_{\text{track}}/10\text{m}, 0, 1)$ and S_{prog} is positive progress divided by nominal progress, capped at one. Continuation feasibility requires no crash, $\nu \leq 0.02$, and $R \geq 0.70$. Preserved tasks require selecting continuation in those six cases. Peaks are maximum absolute roll/pitch and vertical speed; means average selected branches. Counts refer to scenarios, not training replicates.

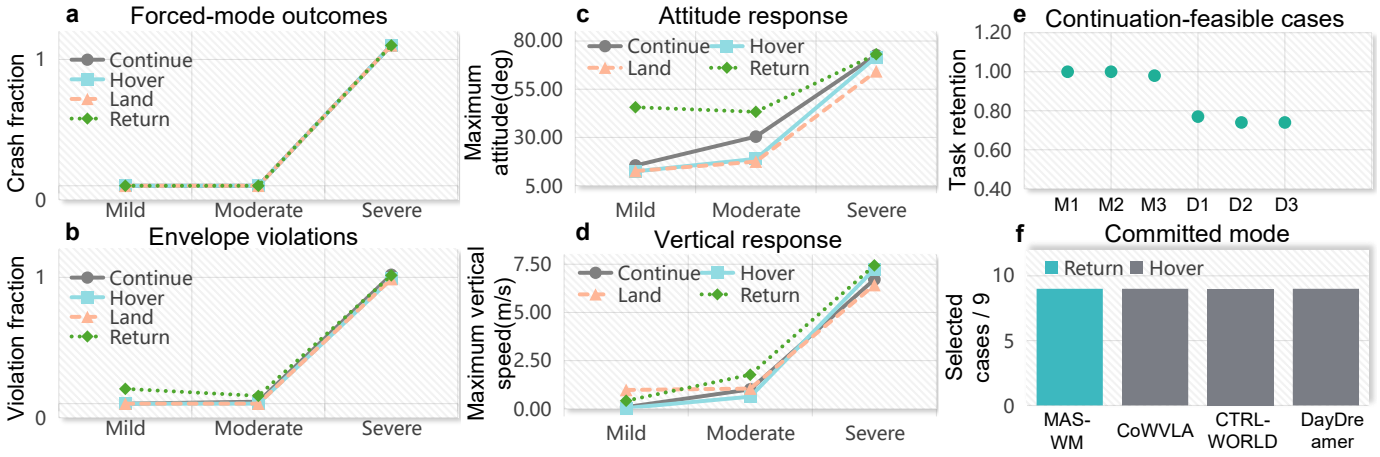


Fig. 9. Controlled motor-fault responses. Motor-3 efficiencies 0.9/0.6/0.3 (severity 0.1/0.4/0.7), three seeds each, give nine scenarios and forty-five trajectories (nominal plus four forced modes). Initial-state differences are at most 0.646 m, 0.079 m/s, and 4.068°, within prescribed tolerances 0.75 m, 0.75 m/s, and 7.5°. (a)-(d) Mode outcomes; coincident crash counts overlap in (a). (e) Retention in six continuation-feasible cases. (f) Choices evaluated in Table V.

1) Application protocol: With the WM frozen, actor $\pi_\eta(a \mid \mathbf{s})$ and critic $V_\xi(\mathbf{s})$ learn from imagined transitions across four modes. For candidate action a, the prior predicts successor $\mathbf{s}'_a$ and reward $\widehat{r}$, yielding

$$Q(\mathbf{s}, a) = \widehat{r}(\mathbf{s}'_a) + \gamma V_\xi(\mathbf{s}'_a),$$
$$\mathcal{L}_\pi = -\mathbb{E}\left[\sum_{a \in \mathcal{A}} \pi_\eta(a \mid \mathbf{s}) Q(\mathbf{s}, a) + \epsilon \mathcal{H}(\pi_\eta)\right]. \tag{16}$$

Here, γ discounts future value, ϵ weights policy entropy, and η, ξ parameterize the actor and critic. The critic minimizes error against the continuation-weighted λ-return:

$$G_t^\lambda = \widehat{r}_t + \gamma \widehat{c}_t \left[(1-\lambda) V_\xi(\mathbf{s}_{t+1}) + \lambda G_{t+1}^\lambda\right]. \tag{17}$$

$\lambda \in [0, 1]$ controls multi-step bootstrapping, with terminal value provided by the critic. Continuation $\widehat{c}_t$ weights Eq. (17). Batch transitions use the modal maximizing action, whereas rewards and continuations use individual maximizers; these assignments may differ. The mode selected upon fault input remains fixed, while subsequent outputs reflect evolving states.

Let $\boldsymbol{u}_t = [u_p, u_\phi, u_v]^\top$ denote clipped, normalized tracking error, mean absolute roll/pitch, and absolute vertical-speed deviation. Their task-deviation mixing weight is

$$b_t = \text{sigmoid}\{\kappa(\boldsymbol{\omega}^\top \boldsymbol{u}_t - \delta)\}, \tag{18}$$

where $\kappa > 0$ controls steepness, $\boldsymbol{\omega}$ contains nonnegative normalized weights, and δ sets the threshold. The reward is

$$r_t = (1 - b_t) r_t^{\text{task}} + b_t r_t^{\text{dev}} + \rho_0 (1 - b_t) + r_t^{\text{term}},$$
$$r_t^{\text{task}} = \rho_p (1 - u_p) + \rho_v \,\text{clip}(\mathbf{v}_t^\top \mathbf{t}_t, -1, 1) + \rho_P \Delta P_t, \tag{19}$$
$$r_t^{\text{dev}} = -\,\text{clip}\left(\frac{\bar{\phi}_t - \phi_0}{\phi_s}, 0, 1\right) - \rho_z u_v.$$

Here, $\mathbf{t}_t$ is the unit task tangent, ΔP_t the progress increment, and $\bar{\phi}_t$ the mean absolute roll/pitch; ϕ_0, ϕ_s define the attitude threshold and scale, ρ coefficients weight reward terms, and r_t^{term} scores crash, landing, or completion. Each actor-critic undergoes 50,000 updates with $H = 15$, $\gamma = 0.99$, $\lambda = 0.95$, $\epsilon = 0.05$, and actor/critic learning rates of $8 \times 10^{-5}/4 \times 10^{-5}$. For Eqs. (18)-(19), $\kappa = 8$, $\boldsymbol{\omega} = (0.5, 0.3, 0.2)^\top$, $\delta = 0.4$, and $(\rho_0, \rho_p, \rho_v, \rho_P, \rho_z) = (0.1, 3.5, 0.8, 20, 0.8)$. Tracking, attitude, and vertical-speed scales are 10 m, 30°, and 2 m/s, respectively; $\phi_0 = \phi_s = 30°$. Terminal rewards are -20, $+5$, and $+15$ for crash, landing, and task completion, respectively.

2) Mode distributions and state-dependent appraisal: Fig. 8 and Table IV characterize actor-critic mode preferences across severity bins. `MAS-WM` maintains non-negligible probability across multiple modes, with decreasing entropy and AAC declining from four to three as severity increases. In

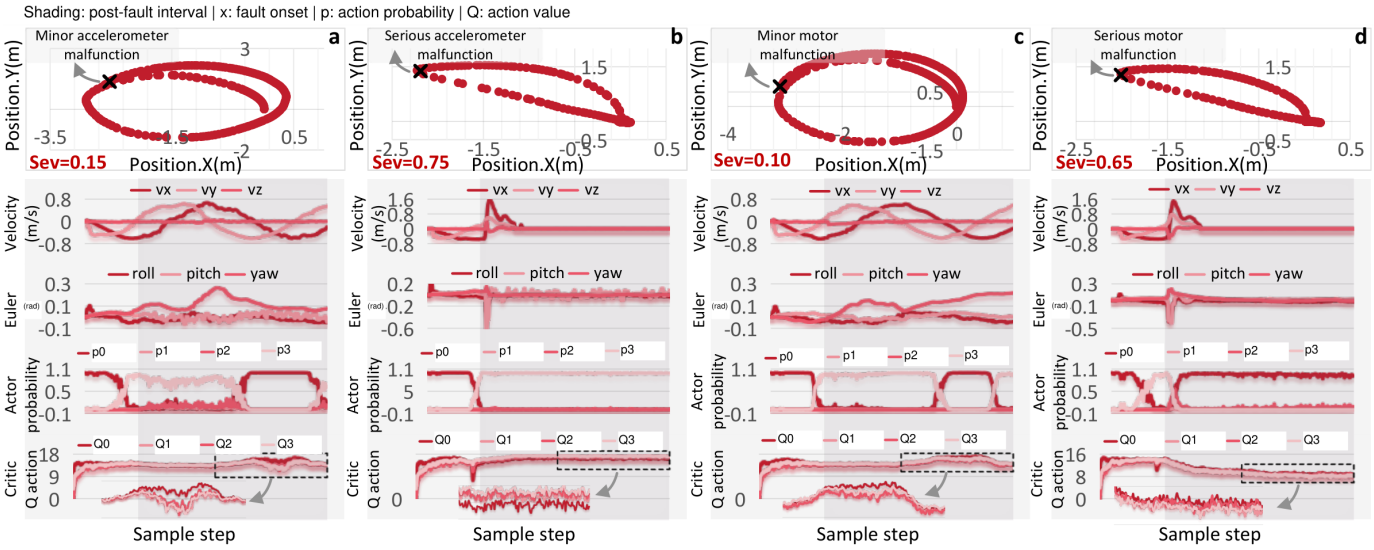


Fig. 10. Flight-state and mode-appraisal records under mild/severe accelerometer and motor conditions. Fault markers and shaded periods align motion with probability/value changes. Angles use rad (tables: degrees); action indices 0-3 denote continue, hover, land, and return. One-step values follow Eq. (16); the selected mode does not change with subsequent probabilities.

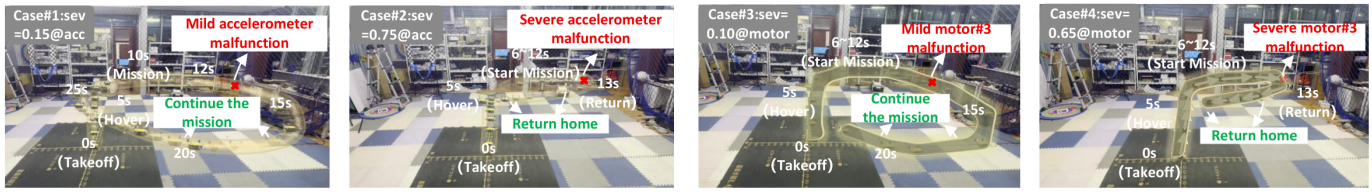


Fig. 11. FS-J150 flight scenes corresponding to Fig. 10: takeoff, hover, task flight, and continuation/return under the four illustrated perturbation conditions.

TABLE VI
COMPLETE EIGHT-TRIAL FLIGHT INVENTORY.

Trial	Condition	Mode	Max. roll (°)	Max. pitch (°)	Max. $\lvert v_z \rvert$ (m/s)	Latency (ms)
T01	Accel. 0.15	Continue	9.29	3.66	0.048	58.32
T02	Accel. 0.15	Continue	11.86	3.82	0.056	59.92
T03	Accel. 0.15	Continue	5.89	3.58	0.049	60.37
T04	Accel. 0.75	Return	14.31	29.17	0.254	61.26
T05	Accel. 0.75	Return	17.09	34.99	0.340	62.97
T06	Motor 0.10	Continue	5.40	2.65	0.107	61.36
T07	Motor 0.65	Return	21.23	22.81	0.512	60.33
T08	Motor 0.65	Return	25.76	39.23	0.480	57.67

Note: Each record has 300 samples at nominal 20 Hz (15 s). Condition is externally supplied and mode is selected with flight-specific guidance. Motion values are absolute window peaks. Latency spans software fault-flag detection to final mode selection, including environment/posterior/selection processing.

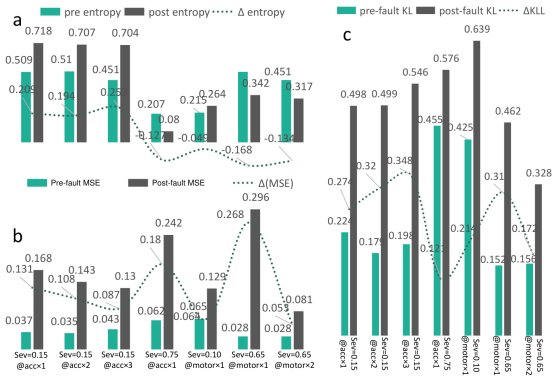


Fig. 12. Model changes for the seven-trial subset in Table VII: pre/post-fault (a) actor entropy, (b) reconstruction MSE, and (c) posterior-prior KL. Dotted curves show post-minus-pre changes; panels distinguish policy concentration from model fit.

contrast, CoWVLA and CTRL-WORLD favor hover, whereas DayDreamer shifts toward return in the highest bin. Under the specified reward and entropy objective, MAS-WM therefore supports a broader distribution of responses across flight states.

Phase-wise statistics distinguish initial mode selection from subsequent state-dependent appraisal. Fig. 8(d) aggregates decisions across states, showing a shift from hover at lower severity to continuation at higher severity, whereas panel (f) isolates initial selection, where return probability reaches 0.9261-0.9676. During execution, per-state entropy increases from 0.245 to 1.148 bits, while continuation probability is 0.3093 (Table IV). Although the selected mode remains fixed, its predicted consequences are continually reassessed as the state evolves, separating initial mode commitment from ongoing response evaluation.

3) Controlled responses and task retention: ***Task retention.*** Fig. 9 compares forced modes from approximately matched states, while Table V reports each policy's selected branch. In six mild/moderate cases, continuation retains 0.738-1.000 of nominal task score without crashes and with violation fractions $\leq$ 0.02. MAS-WM selects return, whereas baselines select hover, sacrificing tasks that remain feasible under continuation.

TABLE VII
PRE/POST-FAULT MOTION IN THE SEVEN-TRIAL DIAGNOSTIC SUBSET.

Trial	Roll pre (°)	Roll post (°)	Δ	Pitch pre (°)	Pitch post (°)	Δ	$\lvert v_z \rvert$ pre (m/s)	$\lvert v_z \rvert$ post (m/s)	Δ
T01	1.08	2.49	+1.41	1.29	1.47	+0.18	0.013	0.019	+0.006
T02	0.94	2.71	+1.77	1.50	1.18	−0.32	0.014	0.023	+0.009
T03	1.53	1.63	+0.10	1.69	1.02	−0.67	0.014	0.014	0.000
T04	1.23	2.03	+0.80	1.69	5.07	+3.38	0.019	0.031	+0.012
T06	1.28	2.17	+0.89	1.68	0.81	−0.87	0.017	0.064	+0.047
T07	1.21	2.61	+1.40	1.85	4.23	+2.38	0.009	0.083	+0.074
T08	1.07	2.50	+1.43	2.01	3.87	+1.86	0.007	0.083	+0.076

Note: Window means of absolute roll/pitch (°) and vertical speed (m/s). Δ is post minus pre. These within-trial averages complement its peaks and the model changes in Fig. 12.

This divergence highlights the distinction between learned response preference and residual task capability.

Maneuver demands. Return requires reorientation and translation, whereas hover prioritizes position holding. Under mild/moderate faults, return induces larger attitude excursions (Fig. 9(c)); across selected branches, its mean violation fraction reaches `0.358`, versus `0.296` for hover, with higher attitude and vertical-speed peaks (Table V). Contingency selection therefore does not necessarily reduce flight deviations. All four modes crash in the three severe cases, indicating that the tested response set cannot prevent failure at this degradation level.

Safety-oriented appraisal. Reward-based imagination trades task progress against flight deviations, a balance made more sensitive to maneuver demands under degraded actuation. The reward in Eq. (19) and trajectory-level retention and exceedance metrics capture complementary response properties; prediction and policy-learning errors further influence mode preference. Joint position, attitude, and velocity predictions enable assessment of these consequences. Incorporating mode-conditioned retention and exceedance estimates into constrained selection could complement return maximization and explicitly account for task feasibility and motion costs.

F. Real-flight validation

1) Setup: Flight trials assess fault-conditioned state and response evaluation on an FS-J150 using the RflySim fault interface. WM inference and policy computation run on a Windows host with an NVIDIA RTX 3080, communicating with PX4 via UDP MAVLink. Faults comprise accelerometer x-axis noise ($\sigma = 3.0/7.5$ m/s^2, labels 0.15/0.75) and motor-3 efficiency degradation (0.95/0.80, labels 0.10/0.65). The observation/action interface matches simulation, while flight uses separate WM/policy parameters and severity-dependent guidance; PX4 provides low-level stabilization.

2) Results: Fig. 10 aligns fault onset with motion, mode probabilities, and one-step values. Mild faults retain task flight, whereas severe faults induce return and larger attitude/velocity excursions. Probabilities and values continue to evolve despite fixed mode selection. Fig. 11 contextualizes these records across takeoff, hover, task flight, and response, illustrating appraisal during physical execution. Table VI summarizes four mild continuation and four severe return trials under the specified guidance. Mild-trial pitch peaks remain below 3.82°, versus 22.81°-39.23° under severe faults, which also produce larger vertical-speed peaks. These responses reflect both fault effects and maneuver demands, motivating joint state prediction. No crashes occur in the eight 15-s records. Table VII separates transient peaks from sustained deviations. Mean absolute roll increases in all seven trials; pitch decreases in T02/T03/T06 but increases in the others, while vertical speed remains unchanged or rises. In T08, for example, the pitch peak reaches 39.23°, despite a post-fault mean of 3.87°. Thus, modest average deviations can mask pronounced transients, underscoring the value of attitude and velocity trajectories alongside position. Fig. 12 distinguishes state inference from response preference. Reconstruction MSE and posterior-prior KL increase after every fault, indicating greater inconsistency with the learned observation and transition models. Entropy rises in mild accelerometer trials but falls in the remaining cases; increased state mismatch can therefore accompany either more diffuse or more concentrated mode probabilities. ***Jointly monitoring state residuals and mode values reveals response evolution beyond policy concentration alone.***

Overall, the deployed model supports continued state-dependent appraisal during fault-affected flight: fault inputs specify operating conditions, observations update the inferred state, and mode probabilities and values evolve accordingly. These trials demonstrate predictive-model deployment in the specified flight system, complementing controlled comparisons of task retention and maneuver demands across modes.

IV. CONCLUSIONS AND FUTURE WORK

MAS-WM connects fault-conditioned state inference to observation-free prediction through a shared-parameter reference-posterior objective. Matched experiments show that retrieval improves reconstruction and reference-posterior regularization further improves multi-step prediction (RQ1). Conditional rollouts preserve useful position and attitude information across the tested faults (RQ2). Downstream evaluation demonstrates state-dependent mode appraisal while exposing conflicts between intervention preferences, task retention, and flight deviations (RQ3). These findings provide an initial exploration of world models for post-fault UAV operation. Future work will treat fault conditions as unknown rather than supplied inputs, integrating fault diagnosis, consequence prediction, and safety-oriented emergency decision-making into a closed loop. This integration will require decisions to adapt to evolving diagnostic confidence and predicted safety outcomes, supported by broader fault conditions, independent test data, and repeated training.

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